# NEXUS SOFTWARE DEVELOPMENT & CODE GUIDELINES

**Category: Engineering | Version: 3.1 | Architecture Board Approved**

## 1. Git Branching Strategy & Pull Request Workflow

Engineers branch from 'main' using convention 'feat/TICKET-description' or 'fix/TICKET-description'. Direct commits to 'main' or 'staging' branches are locked and restricted by branch protection rules. Pull requests require green CI pipeline status, automated linting passing, and 2 approvals.

## 2. API Design & Versioning Standards

All new REST APIs must adhere to OpenAPI 3.0 specifications, return standard error envelopes, enforce token-bucket rate limiting (100 req/min per user), and support semantic versioning.

## 3. Vector Search & Machine Learning Pipeline Guidelines

All semantic vector embeddings must be normalized with L2 norm prior to FAISS index insertion. Chunk sizes must be between 300 and 800 characters with 15-20% overlap to preserve context. Search queries must record latency, similarity scores, and execution metrics for quality auditing.